Lifelong Opportunity Guide 97 — Mechanical Engineering Technician

Version: 2.0 controlled working master
Language: English
U.S. occupation benchmark: O*NET-SOC 17-3027.00 — Mechanical Engineering Technologists and Technicians
Canada comparison: NOC 22301 — Mechanical engineering technologists and technicians
Colombia comparison: CUOC 31150 — Técnicos en ingeniería mecánica
Review date: 2026-08-22

What this career is

Mechanical engineering technicians and technologists help turn engineering ideas into working parts, machines, tests, records, and production improvements. They may support engineers with drawings, prototypes, measurements, equipment setup, testing, troubleshooting, maintenance, manufacturing, quality control, failure analysis, and technical documentation.

The exact title varies. Employers may use names such as mechanical engineering technician, mechanical technician, engineering technician, manufacturing engineering technician, process technician, engineering laboratory technician, mechanical designer, test technician, reliability technician, or engineering technologist.

This guide uses **O*NET-SOC 17-3027.00 — Mechanical Engineering Technologists and Technicians** as the primary U.S. benchmark. It uses **NOC 22301** for the closest Canadian comparison and **CUOC 31150** for the Colombian comparison. These classifications overlap but are not perfectly identical.

The role is not the same as being a licensed professional engineer. A technician may perform sophisticated technical work, but authority to approve or sign engineering designs, regulated calculations, safety-critical changes, or compliance decisions depends on the employer, jurisdiction, project, credential, and delegated responsibility.

Why this role can be an opportunity

Mechanical engineering technology combines practical problem solving with mathematics, tools, measurement, design software, testing, documentation, and teamwork. It can suit people who enjoy understanding how things work and improving them systematically.

Transferable backgrounds can include:

- industrial maintenance;
- machining or CNC work;
- automotive or diesel service;
- HVAC or refrigeration;
- welding or fabrication;
- quality inspection;
- CAD or drafting;
- electronics or mechatronics;
- production operations;
- laboratory work;
- technical military experience;
- manufacturing supervision;
- field service.

A learner does not need to master every mechanical specialty. The stronger strategy is to build a dependable core: read drawings, measure accurately, document changes, work safely, understand tolerances, analyze test results, communicate clearly, and know when to escalate to an engineer or other qualified professional.

What the work can include

Depending on the employer, a mechanical engineering technician may:

- read engineering drawings, sketches, specifications, work instructions, and bills of materials;
- prepare or revise CAD drawings under approved change control;
- create layouts for machinery, assemblies, test fixtures, or production processes;
- assemble and disassemble mechanical systems;
- set up prototypes or test apparatus;
- select and connect approved measurement instruments;
- run tests according to written procedures;
- record dimensions, forces, temperatures, pressures, vibration, speed, flow, or other measurements;
- compare results with engineering specifications and acceptance criteria;
- identify discrepancies and document them accurately;
- support failure analysis and corrective-action investigations;
- prepare test reports, photographs, charts, and technical notes;
- estimate materials, labor, or equipment needs within assigned responsibility;

- prepare work orders or purchase requests;
- support fabrication, machining, assembly, or manufacturing processes;
- help troubleshoot machinery or prototypes;
- support preventive or predictive maintenance programs;
- maintain inspection, calibration, reliability, or maintenance records;
- help coordinate controlled engineering changes;
- verify that the current approved revision of a drawing or procedure is being used;
- support quality inspections and nonconformance documentation;
- assist engineers with research, development, product testing, or process improvement;
- communicate findings to engineers, supervisors, operators, quality personnel, vendors, or maintenance teams.

Technician, technologist, engineer, and mechanic are not the same role

Titles vary by employer and country, but the distinction matters.

Mechanical technician

A technician may focus on hands-on testing, assembly, troubleshooting, measurement, maintenance support, prototype work, documentation, and manufacturing assistance.

Mechanical engineering technologist

A technologist may have broader responsibility for CAD, design support, test planning, process improvement, project coordination, technical calculations, or supervising technical work, depending on education and employer structure.

Mechanical engineer

A mechanical engineer typically has broader responsibility for engineering analysis and design and may hold professional licensure or registration where required. The engineer may own design decisions, calculations, specifications, risk evaluations, and approvals that are outside a technician's authority.

Industrial mechanic or maintenance mechanic

A mechanic may focus more directly on installation, maintenance, disassembly, repair, alignment, bearings, drives, pumps, compressors, conveyors, and production machinery. There can be significant overlap with technician work, but the job classifications, training, and legal scope can differ.

Always follow the actual employer job description and local rules rather than assuming authority from a title.

Core technical skills

Engineering drawings and specifications

Learn to interpret:

- orthographic views;
- sections and details;
- dimensions and tolerances;
- fits and clearances;
- surface-finish callouts;
- materials;
- fasteners;
- weld symbols where relevant;
- notes and revision blocks;
- bills of materials;
- assembly drawings;
- basic geometric dimensioning and tolerancing concepts where required.

Do not change a controlled drawing simply because a field condition seems obvious. Record the discrepancy and use the employer's engineering-change process.

Measurement and metrology

Common tools can include:

- steel rules and tape measures;
- calipers;
- micrometers;
- dial indicators;
- height gauges;
- torque tools;
- pressure gauges;
- temperature sensors;
- tachometers;
- vibration instruments;
- load cells;
- flow instruments;
- data acquisition systems;
- coordinate-measuring equipment in some workplaces.

A measurement is only useful when the instrument is suitable, used correctly, within calibration requirements, and recorded with the correct units and context.

CAD and digital engineering tools

Employers may use:

- AutoCAD;
- SolidWorks;
- Autodesk Inventor;
- CATIA;
- Creo;
- Siemens NX;
- Fusion;
- other 2D or 3D CAD tools;
- product lifecycle management systems;
- document-control systems;
- spreadsheets;
- maintenance-management systems;
- data-analysis or test software.

A specific product name is an employer preference, not a universal legal credential. Learn transferable concepts: sketch constraints, assemblies, revisions, layers, dimensions, drawing standards, file control, configuration management, exports, and documentation.

Testing and data

A technician should be able to:

1. identify the approved test objective;
2. read the procedure and acceptance criteria;
3. verify the correct specimen, assembly, software/firmware version, and drawing revision;
4. confirm instruments and calibration status;
5. identify safety controls and stop-work conditions;
6. record environmental or setup conditions where required;
7. run the test without changing the procedure unless authorized;
8. capture data accurately;
9. document anomalies rather than hiding them;
10. compare results with approved criteria;
11. escalate unexpected or unsafe conditions;
12. preserve traceable records.

Materials and manufacturing awareness

A mechanical technician benefits from understanding the practical behavior of common materials and processes, including:

- carbon and stainless steels;
- aluminum alloys;
- copper alloys;
- engineering plastics;
- elastomers;
- composites where applicable;
- casting;
- forging;
- machining;
- sheet-metal forming;
- welding and joining;
- heat treatment;
- surface coatings;
- additive manufacturing;
- injection molding;
- assembly processes.

You do not need to become a materials engineer to start. You do need to respect approved material specifications and avoid substituting material, fasteners, lubricants, seals, adhesives, or heat-treatment conditions without authorized engineering review.

Maintenance and reliability awareness

Some roles overlap with maintenance and reliability. Useful concepts include:

- preventive maintenance;
- condition-based maintenance;
- lubrication;
- bearings;
- shafts and couplings;
- alignment;
- belts and chains;
- gears;
- pumps and compressors;
- seals and gaskets;
- hydraulic and pneumatic systems;
- vibration;

- wear;
- corrosion;
- root-cause analysis;
- mean time between failures;
- spare-parts control;
- maintenance records.

Do not use reliability concepts to justify continued operation of equipment that has exceeded a safety limit or requires an authorized shutdown.

Professional and engineering boundaries

A technician should not assume authority to:

- sign or seal engineering documents where a licensed or registered professional is required;
- approve safety-critical design changes outside delegated authority;
- select substitute materials for regulated or critical parts without approval;
- change pressure ratings, load ratings, speed limits, temperature limits, torque requirements, guarding, interlocks, or emergency-stop logic without authorization;
- certify that machinery complies with law or standards unless formally authorized and qualified;
- bypass quality hold points;
- approve a nonconforming product for use when disposition authority belongs elsewhere;
- modify production PLC/controller logic, firmware, safety systems, or network settings without explicit authorization and change control;
- perform electrical, welding, pressure, lifting, confined-space, or other specialized work without required training and authorization;
- tell others to enter a hazardous area or defeat a guard because a test is urgent;
- use an AI system as the final authority for engineering calculations, tolerances, safety limits, or compliance decisions.

When scope is unclear, stop and ask the assigned engineer, supervisor, safety lead, quality representative, or other authorized professional.

Safety: hazardous energy and lockout/tagout

Mechanical systems can contain dangerous energy even after normal controls are turned off.

Hazardous energy can include:

- electrical energy;

- rotating mechanical energy;
- compressed air;
- hydraulic pressure;
- steam or process pressure;
- springs;
- gravity from elevated parts;
- thermal energy;
- chemical energy;
- flywheels or stored rotational energy;
- capacitors or other stored electrical energy.

OSHA describes lockout/tagout as a system for controlling hazardous energy during servicing or maintenance. The central lesson is practical: **a stop button or software command is not proof of zero energy**.

Follow the employer's written energy-control program. Do not improvise lockout/tagout. Only trained and authorized people should perform tasks assigned to authorized employees under the applicable program.

Before hands-on work around hazardous machinery, the responsible team may need to:

- identify all energy sources;
- shut down equipment using the approved procedure;
- isolate energy sources;
- apply required locks/tags;
- control stored or reaccumulating energy;
- verify isolation using the approved method;
- maintain control through the work;
- restore equipment using the approved release procedure.

The exact legal requirements depend on jurisdiction and workplace.

Machine guarding and rotating equipment

Mechanical technicians may work near:

- shafts;
- couplings;
- belts;
- pulleys;
- gears;
- chains;
- fans;
- cutting tools;

- presses;
- conveyors;
- rollers;
- robotic cells;
- pinch points;
- hot surfaces.

Never casually remove, bypass, wedge open, tape down, or defeat guards or interlocks. If a test requires an unusual configuration, it needs an approved test plan and controls established by qualified personnel.

Loose clothing, jewelry, hair, gloves, rags, and hand tools can become entanglement hazards around rotating equipment. Follow the employer's PPE, guarding, housekeeping, and safe-distance rules.

Pressure systems and stored mechanical force

Pressure and stored force can release suddenly.

Examples include:

- compressed-air systems;
- hydraulic accumulators;
- compressed springs;
- pressurized vessels;
- piping;
- gas cylinders;
- actuators;
- tensioned belts or chains;
- elevated machine components;
- counterweights.

Never loosen a fitting, remove a cover, open a vessel, release a spring, or work beneath an elevated load based only on an assumption that the system is depressurized or secure. Follow the approved isolation and verification procedure.

Lifting, rigging, welding, and hot work

Some mechanical roles involve cranes, hoists, slings, forklifts, jacks, welding, brazing, cutting, grinding, or other high-energy work.

These activities may require specialized training, inspection, permits, fire-prevention controls, qualified operators, designated riggers, load ratings, ventilation, PPE, and other safeguards. A technician should not accept a high-risk task simply because they understand the mechanical design.

Cybersecurity for mechanical and industrial work

Modern mechanical systems are increasingly connected to digital systems. A technician may encounter:

- CAD/PDM/PLM platforms;
- computerized maintenance systems;
- data loggers;
- industrial PCs;
- programmable controllers;
- networked test equipment;
- sensors and gateways;
- remote-support tools;
- cloud dashboards;
- USB/removable media;
- vendor laptops.

Practical cybersecurity habits include:

- use employer-approved accounts and devices;
- enable MFA when supported;
- use unique credentials and approved password-management tools;
- do not share passwords or access tokens;
- do not plug unknown removable media into industrial or test systems;
- verify unusual requests to install software or change firmware;
- follow approved backup and change-control procedures;
- report suspected phishing, malware, account compromise, or unexplained equipment behavior;
- avoid connecting unapproved devices to production or control networks;
- protect drawings, test data, customer information, and proprietary designs.

Cybersecurity is part of reliability and safety when digital systems influence physical equipment.

Responsible AI in mechanical engineering technology

AI can help with low-risk learning and drafting when the employer permits it.

Reasonable examples include:

- explaining a public technical concept in simpler language;
- creating fictional practice questions;
- brainstorming checklist categories;

- drafting a non-sensitive test-report outline;
- improving grammar in non-confidential documentation;
- suggesting possible failure-analysis questions for human review;
- helping organize public-source research.

Human verification is required for consequential technical work.

Do **not** rely on AI alone for:

- dimensions;
- tolerances;
- loads;
- stresses;
- material selection;
- pressure or temperature limits;
- torque values;
- safety factors;
- lifting limits;
- maintenance intervals;
- test acceptance criteria;
- machine-guarding decisions;
- lockout/tagout procedures;
- engineering changes;
- regulatory requirements;
- final failure conclusions;
- production release decisions.

Do not upload proprietary drawings, confidential specifications, customer information, passwords, access tokens, export-controlled data, private failure records, or production-system information to an unapproved AI service.

A useful rule is: **AI may help draft or explain; approved engineering sources and authorized humans control consequential decisions.**

Education and entry pathways — United States

BLS reports that mechanical engineering technologists and technicians typically need an **associate's degree or other postsecondary training**.

Possible education routes include:

- associate degree in mechanical engineering technology;
- engineering technology certificate;
- manufacturing technology;
- mechatronics;

- CAD/design technology;
- machining or CNC programs;
- industrial maintenance;
- quality technology;
- automation technology;
- employer-sponsored technical training.

Before paying for a program, compare:

- total tuition and fees;
- tool or software costs;
- accreditation where relevant to your goals;
- laboratory access;
- CAD instruction;
- metrology and testing content;
- internship or co-op opportunities;
- employer partnerships;
- job-placement evidence;
- transfer options to a bachelor's degree if desired.

Accreditation can matter for some education or certification pathways, but this guide does not certify or accredit any program.

U.S. funding and lower-cost training

WIOA and American Job Centers

CareerOneStop provides a WIOA-Eligible Training Program Finder:

https://www.careeronestop.org/LocalHelp/EmploymentAndTraining/find-WIOA-training-programs.aspx

If eligible, a learner may be able to access approved training support. Eligibility, program approval, supportive services, and available funding vary by location and individual circumstances.

Search related categories, not only the exact title:

- mechanical engineering technology;
- manufacturing technology;
- mechatronics;
- industrial maintenance;
- CAD;
- CNC machining;
- quality technology;
- automation;

- engineering technician.

An American Job Center can explain current local options.

Work-based learning

Formal Registered Apprenticeship availability varies. Do not assume a mechanical engineering technician apprenticeship exists in every location.

Other practical routes may include:

- paid technician trainee roles;
- manufacturing apprenticeships;
- industrial maintenance apprenticeships;
- employer-sponsored community-college programs;
- internships;
- co-op education;
- paid laboratory or test work;
- tuition reimbursement;
- structured vendor training;
- supervised prototype or production assignments.

Canada pathway

Canada Job Bank classifies the occupation under **NOC 22301 — Mechanical engineering technologists and technicians**.

Typical requirements include:

- a two- or three-year mechanical engineering technology college program for technologists;
- a one- or two-year mechanical engineering technology college program for technicians;
- supervised experience before some certifications;
- provincial certification that may be requested for some jobs;
- in Quebec, regulated use of the title Professional Technologist.

Always check the current provincial requirements for the exact title and role.

Canada wages and outlook

Job Bank's national wage data, updated November 19, 2025, reports:

- low: **C$23.08/hour**;
- median: **C$35.00/hour**;
- high: **C$51.28/hour**.

These are national figures and vary by province, industry, experience, certification, and specialty.

Job Bank reports a **moderate risk of labour shortage nationally over 2024–2033**, while near-term prospects differ across provinces and territories.

Canada training and funding supports

Start with:

https://www.canada.ca/en/services/jobs/training.html

Canada's Labour Market Development Agreements and Workforce Development Agreements support training, upskilling, employment services, work experience, and in some programs wage subsidies or employer-supported training. Eligibility varies.

Colombia pathway

Colombia's OCUPACOL classifies **CUOC 31150 — Técnicos en ingeniería mecánica** at competence level 3.

A strong Colombian pathway can include SENA technical/technologist training, manufacturing experience, maintenance experience, CAD, metrology, industrial safety, and employer-specific equipment training.

SENA examples

SENA Betowa lists **Mantenimiento Mecánico Industrial** as a Tecnólogo program of **3,984 hours**. Its competencies include industrial equipment installation, preventive maintenance, repair, technical procedures/manuals, safety and environmental practices, quantitative reasoning, English, and digital tools.

SENA also lists **Mantenimiento Electromecánico Industrial** as a Tecnólogo pathway of **3,984 hours**, combining mechanical maintenance with electrical-machine and control-system work.

Program availability changes by city, cohort, date, modality, and employer partnership. Admission is subject to available seats and SENA's selection process. Verify the live Betowa listing before making plans.

Latin America and Caribbean pathway

OIT/Cinterfor can help identify national vocational-training institutions across Latin America and the Caribbean:

https://www.oitcinterfor.org/

https://www.oitcinterfor.org/statsfp/paises

Use it as a locator. Search national institutions for programs in:

- mechanical technology;
- industrial maintenance;
- manufacturing;
- mechatronics;
- CAD;
- machining;
- quality;
- automation;
- welding/fabrication;
- reliability.

Verify current cost, admission, modality, credential, funding, and employer recognition directly with the national institution.

Income research — official sources first

United States official benchmark

O*NET reports **2025 median wages of $35.82/hour and $74,510/year** for SOC 17-3027.00, using BLS 2025 wage data.

The BLS Occupational Outlook Handbook reports an earlier **May 2024 median of $68,730/year**. Use the newer O*NET/BLS 2025 median as the principal current wage benchmark while keeping the OOH value clearly dated.

Canada official benchmark

Job Bank reports **C$23.08 low / C$35.00 median / C$51.28 high per hour**, updated November 19, 2025, for NOC 22301.

Colombia official classification caution

OCUPACOL provides classification and labour-market indicators, but its displayed salary information should not be treated as the main current wage benchmark without confirming the underlying reference period and methodology. Use it primarily for Colombian occupational classification and role context.

Current non-government market estimates

Private salary sites can be useful only as secondary context because their methods and title definitions differ.

United States — ZipRecruiter

As of August 7, 2026, ZipRecruiter reported about **$75,124/year ($36.12/hour)** for the title Mechanical Engineering Technician.

Source: https://www.ziprecruiter.com/Salaries/Mechanical-Engineering-Technician-Salary

United States — Salary.com

As of August 1, 2026, Salary.com reported about **$58,275/year ($28/hour)**, with a 25th–75th percentile range around **$51,667–$65,216**.

Source: https://www.salary.com/research/salary/listing/mechanical-engineering-technician-salary

The difference between private estimates is a reminder that they are not official statistics.

Colombia — broader technician-title reference

Computrabajo's Técnico mecánico page, updated June 27, 2026, showed employer-level averages commonly around roughly **COP 1.4M–1.8M/month** among several displayed employers. This is a broader mechanic/technician title and is **not** a direct equivalent of CUOC 31150.

Source: https://co.computrabajo.com/salarios/tecnico-mecanico

Use current job postings and employer offers to evaluate actual compensation for a specific role.

Accessibility and job fit

Mechanical technician work is not physically identical across employers.

Some positions are mostly:

- CAD;
- laboratory testing;
- quality documentation;
- planning;
- technical writing;
- data analysis.

Others involve:

- standing and walking;
- climbing;
- lifting within assigned limits;
- factory noise;

- PPE;
- field work;
- travel;
- heat or cold;
- shift work;
- machinery exposure.

Read the actual job's essential functions. If you need an accommodation, use the applicable employer or school process. This guide does not determine legal entitlement to any particular accommodation.

For learning accessibility, use:

- short study sessions;
- step-by-step checklists;
- diagrams paired with written explanations;
- consistent units;
- labeled examples;
- spaced repetition;
- speech-to-text or text-to-speech tools where helpful;
- accessible digital notes;
- practice with real or simulated parts only under safe supervision.

Build a beginner portfolio without unsafe claims

A portfolio can show technical thinking without pretending to be a licensed engineer.

Possible projects:

- a dimensioned CAD model of a simple non-safety-critical bracket;
- an assembly drawing with a bill of materials;
- a fictional test plan and test report;
- a measurement study comparing caliper and micrometer readings;
- a maintenance checklist for a fictional training device;
- a simple tolerance-stack practice exercise clearly labeled educational;
- a root-cause exercise using a fictional failed component;
- a spreadsheet that charts fictional temperature or vibration data;
- a revision-control example showing how a drawing change is documented;
- a safe 3D-printed prototype project.

Label educational work clearly. Do not present classroom or simulated work as professional engineering approval.

Six-step action plan

This controlled edition repairs the legacy action-plan defect. Each step has a checkable milestone.

Step 1 — Confirm fit and target role

Actions:

- review at least 10 current technician/technologist job postings;
- note recurring duties, tools, education, shift, physical demands, and safety requirements;
- choose one initial target such as test technician, manufacturing engineering technician, mechanical technician, CAD technician, or maintenance/reliability technician.

Milestone: one-page role comparison completed with a selected target role and three reasons it fits.

Step 2 — Build the technical foundation

Actions:

- refresh algebra, geometry, units, and basic physics;
- learn drawing interpretation, tolerances, materials, measurement, and basic mechanical systems;
- practice unit conversions and dimensional checks.

Milestone: complete a 20-question self-test at 80% or better and correct every missed answer.

Step 3 — Learn one practical tool chain

Actions:

- learn one mainstream CAD platform available to you;
- practice spreadsheets for calculations and test data;
- practice at least two measurement tools in a safe supervised setting if available.

Milestone: produce one dimensioned part drawing, one simple assembly, and one measurement/data worksheet.

Step 4 — Build safety and controlled-work habits

Actions:

- study your training program's machinery and energy-control rules;
- learn why guards, interlocks, LOTO, stored energy, pressure, lifting, and hot work require formal controls;

- practice writing a stop-and-escalate decision for unsafe scenarios.

Milestone: complete a written safety checklist with at least 10 hazards and the correct escalation owner for each.

Step 5 — Gain supervised experience and evidence

Actions:

- pursue an internship, co-op, trainee role, laboratory project, manufacturing assignment, supervised maintenance work, or employer-supported learning;
- keep non-confidential notes on tasks, tools, measurements, problems, and outcomes;
- request feedback on accuracy, safety, documentation, and teamwork.

Milestone: document three supportable accomplishment examples using situation, action, and result without exposing confidential information.

Step 6 — Apply, interview, and improve

Actions:

- tailor the résumé to the target job using only truthful evidence;
- prepare examples of drawing interpretation, measurement, testing, troubleshooting, safety, and teamwork;
- apply consistently to realistic roles;
- track results and revise the approach based on responses.

Milestone: submit 10 well-matched applications, complete two mock interviews, and review the tracker for the next improvement.

Interview preparation

Be ready to explain:

- how you verify the current drawing revision;
- how you handle a measurement outside tolerance;
- why calibration status matters;
- how you document a test anomaly;
- what you do if a machine is unexpectedly energized;
- why you do not bypass a guard to save time;
- how you escalate a design discrepancy;
- how you keep units and calculations consistent;
- how you protect confidential engineering data;
- how you use AI without trusting it for safety-critical facts.

A strong answer is usually specific, careful, and honest about authority boundaries.

Final checklist

Before applying, confirm that you can honestly say:

- ☐ I can explain what a mechanical engineering technician does.
- ☐ I understand the difference between technician support and professional-engineering authority.
- ☐ I can read basic mechanical drawings and identify revision information.
- ☐ I can use at least one measurement tool correctly or explain how I am training to do so.
- ☐ I understand why calibration and traceability matter.
- ☐ I can describe a controlled test process.
- ☐ I understand the purpose of machine guarding and lockout/tagout.
- ☐ I know that stored energy can remain after normal shutdown.
- ☐ I can explain when to stop work and escalate.
- ☐ I can protect engineering data and follow employer cybersecurity controls.
- ☐ I can explain safe, limited use of AI in technical work.
- ☐ I have checked current education and funding options in my location.
- ☐ I have one truthful portfolio or experience example relevant to the target role.
- ☐ I have completed all six action-plan milestones or have dates assigned to finish them.

Sources and verification links

United States

- O*NET OnLine — Mechanical Engineering Technologists and Technicians: https://www.onetonline.org/link/details/17-3027.00
- U.S. Bureau of Labor Statistics — Occupational Outlook Handbook: https://www.bls.gov/ooh/architecture-and-engineering/mechanical-engineering-technicians.htm
- BLS 2025 OEWS national table: https://www.bls.gov/news.release/ocwage.t01.htm
- CareerOneStop WIOA-Eligible Training Program Finder: https://www.careeronestop.org/LocalHelp/EmploymentAndTraining/find-WIOA-training-programs.aspx
- OSHA — Control of Hazardous Energy: https://www.osha.gov/control-hazardous-energy

- OSHA Lockout/Tagout guide: https://www.osha.gov/sites/default/files/publications/OSHA3120.pdf
- CISA Secure Our World: https://www.cisa.gov/secure-our-world
- NIST AI Risk Management Framework: https://www.nist.gov/itl/ai-risk-management-framework
- NIST Generative AI Profile: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence

Canada

- Job Bank summary: https://www.jobbank.gc.ca/marketreport/summary-occupation/3236/ca
- Job Bank description: https://www.jobbank.gc.ca/marketreport/occupation/3236/ca
- Job Bank wages: https://www.jobbank.gc.ca/marketreport/wages-occupation/3236/ca
- Job Bank prospects: https://www.jobbank.gc.ca/marketreport/outlook-occupation/3236/ca
- Job Bank requirements: https://www.jobbank.gc.ca/marketreport/requirements/3235/ca
- Government of Canada training: https://www.canada.ca/en/services/jobs/training.html
- Labour Market Agreements: https://www.canada.ca/en/employment-social-development/programs/training-agreements.html
- Labour Market Development Agreements: https://www.canada.ca/en/employment-social-development/programs/training-agreements/lmda.html

Colombia and Latin America

- OCUPACOL CUOC 31150: https://ocupacol.mintrabajo.gov.co/Profile/OccupationalProfile/31150
- OCUPACOL hierarchy: https://ocupacol.mintrabajo.gov.co/Jerarquias/Jerarquias
- SENA — Mantenimiento Mecánico Industrial: https://betowa.sena.edu.co/oferta/mantenimiento-mecanico-industrial?level=6&modality=P&programId=130255
- SENA — Mantenimiento Electromecánico Industrial: https://betowa.sena.edu.co/oferta/mantenimiento-electromecanico-industrial?modality=P&offertype=company&programId=130695&search=mecanico
- OIT/Cinterfor: https://www.oitcinterfor.org/
- OIT/Cinterfor country network: https://www.oitcinterfor.org/statsfp/paises

Non-government compensation context

- ZipRecruiter: https://www.ziprecruiter.com/Salaries/Mechanical-Engineering-Technician-Salary
- Salary.com: https://www.salary.com/research/salary/listing/mechanical-engineering-technician-salary
- Computrabajo Colombia: https://co.computrabajo.com/salarios/tecnico-mecanico

Important notice

This guide provides general educational and career-planning information. It is not engineering, legal, safety, financial, immigration, employment, or licensing advice. Requirements vary by employer and jurisdiction. Verify consequential decisions through current official sources, employer procedures, equipment manuals, applicable standards, and qualified professionals.

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Author and AI assistance

Created and directed by **Alberto “Al” Leiva**. ChatGPT supported research, organization, editing, translation support, and document preparation under the author's direction. The author remains responsible for editorial and publication decisions.